

Rethinking Lived Experience in Chronic Illness: Navigating Bodily Doubt with Consumer Technology in Atrial Fibrillation Self-Care

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Abstract

Consumer technology is increasingly used to support the self-care of atrial fibrillation (AF), a chronic heart condition that affects physical, emotional, and mental health due to its unpredictability, symptoms, and complications. Through interviews with 29 adults self-tracking while living with AF, we found that consumer technology enabled participants to outsource bodily awareness to their ‘digitised heart,’ facilitating innovative pill-in-pocket interventions and empowering negotiation in shared decision-making. Drawing on phenomenology, we introduce ‘Bodily Doubt’ to explain how uncertainty about the body shapes the use of technology in chronic illness and how the use of technology influences uncertainty. Technology mediates ‘Bodily Doubt’ both by providing reassurance and exacerbating it, particularly when technology fails to adapt to disease progression. Our findings have implications for understanding how technology influences the lived experience of illness, challenging experiential concepts of lived experience in self-tracking and design that foregrounds the experience of the lived body.

CCS Concepts

• **Human-centered computing** → Empirical studies in HCI.

Keywords

Self-Tracking, Cardiovascular Disease, Self-Care Technology

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1 Introduction

Atrial fibrillation (AF) is a cardiac condition affecting 59 million people globally, with its prevalence rising [69, 106]. Cardiologists liken AF to the “*lower back pain*” of cardiology: chronic, common, and notoriously difficult to manage [30]. The impact of AF affects various bodily systems and can cause serious complications such as stroke, heart failure, cognitive decline, and premature death [43, 46, 101, 129]. Managing AF is challenging as AF can occur unexpectedly and bring severe and disabling symptoms [17]. It significantly impacts a patient’s quality of life, physically and mentally [2, 63]. Psychological stress and anxiety are common, affecting between 25% to 45% of AF patients [64, 111, 123].

Self-care is critical to slowing the progression of the disease, but current clinical approaches leave people with AF feeling uninformed and unsupported in their everyday care [79, 127]. Increasingly, people with AF are turning to consumer technologies with medicalised features to support self-care. For example, a survey by the Stop Afib charity revealed that 71% of respondents used consumer technology to manage their AF [49]. Whilst some tools are FDA-approved to diagnose AF, they have not been designed or regulated for self-care [120]. Though the human-computer interaction (HCI) community has explored self-tracking for chronic condition management, research on AF self-care is limited [5, 81, 83, 109, 110]. Much of the existing research predates the development of features such as electrocardiograms (ECGs) and irregular heart rhythm alerts.

The medical community has expressed cautious optimism about these technologies [90, 120]. Recent work shows that patient-generated data (PGD) from consumer technology is increasingly used in cardiology clinical practice [7, 9, 77]. However, significant concern remains regarding their accuracy and their potential to heighten health anxiety [76, 104, 105, 120]. Despite the increasing use of consumer technology by both patients and cardiologists, little research has explored how consumer technologies are used for AF self-care [7, 9].

To address this gap, we conducted semi-structured interviews with 29 people living with AF to understand how they use technology in their self-care. We contribute an empirical account of how consumer devices are used in AF self-care, showing how individuals outsourced bodily awareness by offloading their felt experiences

onto technology. This quantification of felt experiences gives rise to the notion of the 'digitised heart', an externalised representation of their heart health.

We also contribute an account of innovative technology uses, including a novel, technology-mediated approach to a pill-in-pocket intervention. Participants repurposed technology to cast a clinical gaze upon themselves, using self-tracked data to emulate clinical monitoring. Through pre-clinical 'data work', participants constructed their own clinical narrative, which they used in clinical encounters to influence and shape shared decision-making in line with their treatment preferences.

Finally, we document the emotional toll of illness and self-tracking, demonstrating how self-care practices can result in a complex entanglement of emotion and technology.

Drawing on phenomenology, we introduce 'bodily doubt' to explain our findings [20]. 'Bodily doubt' occurs when chronic illness disrupts the implicit taken-for-granted sense of bodily capacity experienced in health [21]. We discuss how 'bodily doubt' drives the use of technology in chronic disease. Technology can mediate bodily doubt by initially providing reassurance, but it can also exacerbate uncertainty as illness progresses and technology fails to adapt. This work challenges the concept of lived experience in models of self-tracking and questions design approaches that foreground the experience of the lived body.

2 Related Work

In this work, we refer to self-care according to Riegel et al.'s theory of self-care: as maintenance, monitoring, and self-care management [99]. For ease of reading, we will use the term self-care as an overarching concept that includes self-management. We now review AF and the role of self-care, followed by the current and emerging role of consumer technology in cardiology. We then outline personal informatics (PI) and self-tracking for chronic conditions in HCI. We define the scope of our work and how it builds on existing HCI research.

2.1 Atrial Fibrillation and The Role of Self-Care

AF is a disease continuum, starting with early risk stages (1-2) where individuals may have electrical or structural heart abnormalities [18, 55]. Surveillance and risk reduction are key in these phases. Stage 3 involves intermittent AF, with treatments like medication, cardioversion (an electrical procedure to reset the heart's rhythm) or ablation (which heats or freezes areas of the heart). Silent AF (no symptoms) can occur in 10-40% of cases, which carries the same risks as symptomatic AF [108]. Early intervention and shared decision-making are crucial for preventing disease progression across stages 1-3, while Stage 4 shifts toward managing symptoms without intervention [55].

Self-care has a critical role in preventing disease progression and serious complications like stroke and heart failure across all AF stages [55, 100, 127]. Practices such as self-monitoring, medication adherence and lifestyle modification are essential for improving health outcomes and quality of life [8, 92, 93, 98, 127]. Lifestyle modification can be particularly impactful in the early risk stages, as it can prevent AF from developing, with evidence also showing

significant benefits such as reversing or slowing AF progression for those already diagnosed [40, 68, 94, 95].

However, unique barriers hinder the adoption of self-care practices among people with AF. Early-risk AF patients receive limited support due to healthcare systems prioritising treatment over prevention [100]. For those with asymptomatic or episodic AF, the transient nature of episodes may diminish motivation to initiate self-care [128]. In contrast, highly symptomatic patients may rely on emergency care rather than engage in self-maintenance strategies [128].

Despite its importance, self-care remains under-utilised. Clinicians and patients receive little training, and short medical appointments limit the opportunity for clinicians to provide the knowledge and skills required for self-care decision-making [100]. A recent systematic review of AF self-care interventions found that delivering even one element of self-care, such as self-monitoring or self-maintenance, resulted in significant health improvements when compared to usual care [127]. When technology was integrated alongside personalised educational interventions, this further enhanced outcomes [127].

The unpredictable nature of the condition and the complexity of an evolving treatment regime means that patients need ongoing self-care support [60, 128]. Yet self-care initiatives are lacking in clinical practice [100]. As a consequence, those living with AF learn to navigate self-care through trial and error, leaving them feeling isolated and ill-equipped to self-manage their condition [79, 127].

2.2 Current and Emerging Role of Consumer Technology in Cardiology Practice

Consumer technologies are becoming more popular, especially for heart health monitoring, offering the potential for early diagnosis, personalised treatment of arrhythmia like AF and improving health outcomes cost-effectively [28, 120, 121]. These technologies are accessible, affordable, and provide indefinite monitoring, though not continuously [120], which is in contrast to prescribed monitoring which is harder to access, expensive, and offers limited monitoring time [120].

Consumer technologies allow people with AF with infrequent AF episodes to capture clinically actionable data [120] that is recognised as evidence by medical organisations [22, 55, 119]. The use of consumer technology for periodic arrhythmia monitoring has been evaluated and shown to be feasible for AF detection in numerous studies [47, 71, 96, 102]. However, much of the evidence generated to date has been in collaboration with device manufacturers and within controlled environments [22, 58]. There remains a significant gap in independent, real-world evidence, particularly in vulnerable populations [22]. Despite growing popularity, many consumer technologies fall outside regulatory oversight [78], which leaves patients and clinicians navigating the divide between 'regulated health' and 'exempt wellness' [84].

Despite lack of evidence and regulation, patient-generated data (PGD) from consumer technologies is increasingly used in cardiology practice [7, 9, 77]. A'Court et al found that in 69% of cardiology outpatient cases, PGD supported a diagnosis, primarily of atrial arrhythmias such as AF [7]. Additionally, in 29% of cases, PGD was used in managing pre-existing cardiovascular conditions [7]. This

was observed beyond AF patients but also in those with hypertension, coronary artery disease, and heart failure. However, in 31% of cases, clinicians did not utilize PGD [7]. Concerns remain over inappropriate referrals [7, 77].

Emerging clinical trials are currently exploring the use of wearables in notifying a patient that an AF episode has started followed by a prompt to take medication in real-time to manage the risk of stroke [91]. AF patients are five times more likely to have a stroke, with 70-80% resulting in death or serious disability [45, 107]. Currently, anti-stroke medications are taken continuously and are expensive and unpopular among patients, as they severely impact quality of life, leading to poor long-term compliance [90]. This targeted approach could remove the need for lifelong daily medication [91]. Doctors can currently offer this off-label approach when deemed appropriate, but is not yet widely adopted, and continuous monitoring algorithms for alerts are only available to trial participants [91].

Despite the potential of consumer devices in AF self-care, challenges remain. There is a lack of clinical guidelines for doctors on how to use, interpret, and act upon PGD [7, 54]. Integration of PGD into electronic health records and reimbursement need addressed [22]. Furthermore, the constant health surveillance wearables offer may lead to heightened anxiety in AF patients [104]. Rosman et al. [105] found that wearable users were significantly more likely to engage in obsessive symptom monitoring, and experience anxiety and fear from device alerts when compared to non-users [105]. The study also revealed higher use of medical resources (e.g. ablations) amongst wearable users [105]. However, wearable users in the study reported a higher burden of symptoms, thus requiring more intervention [22]. More research is needed to determine if healthcare use is linked to wearable use [22].

2.3 Personal Informatics Models Moving Towards Lived Experience

Personal Informatics (PI) models provide a valuable foundation for understanding consumer self-tracking technology and how people experience these tools, which is fundamental for PGD. Li et al. [66] developed a goal-oriented, stage-based model (preparation, collection, integration, reflection, and action). Each phase could be user-driven, system-driven, or a combination, influencing behaviour change [66]. Choe et al. [25] expanded on this by defining data collection (fully manual, fully automated, or semi-automated), highlighting how these collection methods influence reflection and behaviour change.

There was a swing towards the experiential in PI discourse with Rooksby et al. [103] introducing lived informatics, shifting focus from goal-oriented tracking to practices that are "*enmeshed in everyday life*." They identified five styles of tracking: directive, diagnostic, documentary, rewards, and fetishised tracking [103]. This shift to the experiential in PI prompted more customisable and flexible designs [6, 29]. Epstein et al. [35] built on this with the Lived Informatics model that considers emotion, social context and lapses in tracking. Other work has raised the argument that reflection may not always be possible, which risks rumination and abandonment [33]. PI models have evolved to show how each process is multifaceted, such as sense-making and reflection [24, 29]. Researchers

have also gone beyond models using personal data instead to foreground the body's lived experience in design, translating biodata into somadata "*to spur first-person, felt, explorations*" [1].

These PI models assume health and do not consider periods of illness [5, 27]. The self-improvement hypothesis has dominated the design and framing of PI models [57]. This framing assumes that data and self-insight lead to favourable behaviour changes [57, 66]. With PI designs reinforcing the portrayal that health and wellbeing is a personal responsibility [73]. Elsdén et al. [34] aptly describe how PI models and designers often assume that users are tracking to "*analyse their run today in an attempt to run faster tomorrow*." However, for those living with chronic illness such as AF, one might analyse their run today, only to face the unpredictability of tomorrow, when breathlessness and fatigue make yesterday's run not just unrepeatable but impossible to improve upon. For PI users living with a chronic condition, self-improvement is not the primary focus. While PI models are useful for understanding the processes in which patient-generated data would be collected and used for design, they fail to account for the lived experience and mundane work of living with a chronic condition [88], whilst also overlooking the collaborative and social factors that shape health and wellbeing [73].

2.4 Self-Tracking Whilst Living With A Chronic Condition

Self-tracking has been explored by the HCI community as a tool to support self-care in chronic health conditions [5, 81, 83, 88, 109, 110]. From a cardiovascular perspective, the HCI community has explored the role of self-tracking to support cardiac diagnosis, cardiac rehabilitation and the management of heart failure and hypertension [32, 58, 62, 75, 116, 117]. Self-tracking has been shown to assist elements of self-care, such as supporting diagnosis and identifying symptom triggers [56, 58], whilst also fostering a sense of control over illness and encouraging reflection on health conditions [5, 19]. Self-tracked data has also been shown to help patients make more informed health decisions and facilitate collaboration between patients and healthcare providers [26, 53, 80].

However, several barriers can limit engagement with self-tracking in the context of self-care. Sefidgar et al [110] found that current technologies often leave users unsupported in deciding what is important to track and uncertain how to adapt their tracking over the course of their condition. Keys et al. [58] note that the interpretation of complex PGD is often incomplete as collaborative sense-making with a doctor is not always accessible [58].

Some have argued that self-tracking tools need to go beyond quantifying disease [124], with self-tracking failing to consider the everyday effort involved in illness management [88]. For instance, it can be a social process rather than solitary, an aspect often overlooked in self-tracking design [27, 122]. There has been a growing movement towards more holistic 'whole person' design that supports not only physical health but also mental and emotional well-being [5, 27, 48, 65]. With calls for technology to support aspects such as bodily awareness and self-efficacy as part of self-care [27], this perspective contributes to a broader HCI discourse that advocates for less medicalised designs and a stronger focus on

the everyday lived experiences of individuals managing long-term chronic conditions [86].

2.5 Atrial Fibrillation and HCI

There is limited research on living with AF and using technology. Keys et al. explored the use of consumer self-tracking technologies, such as the Apple Watch, to support the diagnosis of cardiovascular conditions, finding that these tools helped bridge gaps in clinical diagnostic technologies [58]. Participants used these tools to confirm symptoms and create an evidence trail for clinical encounters, with several reporting a resulting AF diagnosis [58].

Recent work on AF has focused on designing novel sensors and improving AF detection [67, 130]. Much of the research focus has been placed on supporting clinicians, such as She et al., who co-designed a remote patient monitoring system with decision support using explainable AI to assist cardiologists in managing AF patients post-ablation [112]. Fanio et al. [36] recognised a need for decision aids to help AF patients decide treatment options. However, this was based on the translation of clinical guidelines to the patient audience, and did not consider the role of patient-generated data in potentially creating a personalised treatment approach. They did identify several design challenges specific to this population, including managing patient anxiety while respecting their desire for full access to health data, visualising symptom-based data and quality-of-life measures, and ensuring the tools were accessible to all [36].

In summary, despite how common AF is, the practice of using consumer tools to support self-tracking and self-care has not been explored. Given the growing influence consumer technology is having on cardiology clinical practice and its potential role in guiding medical interventions, it has never been more important to understand how these tools shape an individual's lived experience. This is important for conditions such as AF where a person's lived experience of the condition might be 'silent' or asymptomatic, yet significantly impact health.

This work aims to extend the current literature by answering the following research question:

- How do individuals use and experience consumer technology whilst living with AF?

3 Method

This study examines the lived experience of using consumer technologies while living with AF. To answer our research question, we conducted semi-structured interviews with 29 people with AF using consumer technologies in self-care. Ethical approval was obtained from the University of Bristol Institutional Review Board (IRB).

3.1 Participants

We recruited 29 participants via an AF support group on a popular online forum. Two approaches to recruitment were taken. First, a private message was sent to prospective participants via the forum's online chat function. These prospective participants were contacted if they had shared experiences of using self-tracking tools in the online forum. The private message contained a recruitment message and a link to a Microsoft Form. The form contained further study information, a consent form and a demographic questionnaire.

Secondly, we posted a public recruitment message on the online forum. Prospective participants were encouraged to post a comment if interested. These prospective participants were then sent a private message with the same content as participants in the first phase of recruitment. Nine participants had been part of a prior study using the same online forum. To protect the anonymity of the participants, the online forum used in this study is not disclosed in line with the British Psychological Society's (BPS) approach to internet-mediated research [114].

Participants were aged over 18 years old with self-declared AF and experience using consumer technologies as part of self-care. Participants fell under various public and private health systems, such as the National Health Service (NHS) in the UK, Medicare in the US, and private health providers. Table 1.1 describes participant demographics. Participant ages ranged from 31 to 76 (mean = 52). Most participants identified as male, with 7 female participants. They have diverse experience using various self-tracking technologies, consumer ECG monitors, and third-party applications. Participants were mostly highly educated, with 12 participants with a bachelor's degree and 8 participants with a master's degree or above. Of the participants who declared annual income, 10 had income > \$100,000 and 3 had income > \$200,000.

3.2 Interview Data Collection

To gain deeper insights into participants' lived experiences, we conducted semi-structured interviews. Questions varied based on the participants' stage of AF, as self-tracking practices may shift with the condition's progression. We initially focused on why participants chose to self-track (e.g. "Can you explain what motivated you to self-track whilst managing your condition?" and "Why did you choose this particular technology?"). Next, we explored how participants self-tracked whilst living with AF (e.g. for those with few AF episodes, "How has the use of your smartwatch changed, if at all, since your diagnosis?", for those with more frequent AF episodes "Can you talk me through what happens when you experience an AF episode?", and for those with long periods in AF or permanent AF "Can you talk through your typical use of your device?"). We prompted participants to reflect on how their use of self-tracking technology had evolved throughout the course of their illness. We then explored the experience of self-tracking whilst living with AF (e.g. "Has the data influenced your decision-making regarding your condition?"). We also aimed to identify any benefits or challenges participants may have encountered whilst self-tracking, for example, by inquiring whether there had ever been a time when the technology caused anxiety or if there had been any occasions where the participant felt that they may have been using the technology more than necessary. Additionally, we inquired about the use of self-collected data in clinical encounters (e.g. "How do you decide which data to share with your healthcare provider?").

Interviews lasted between 30-75 minutes, resulting in over 30 hours of audio recordings. Interviews were conducted via Microsoft Teams. Participants were reimbursed for their time via a chosen charitable donation or £20 voucher.

ID	Gender	Age	Location	Primary Tracking Tool	Secondary Tracking Tools	Form of AF	Time Self-Tracking & Self-Managing Condition (years)
P1	Male	32	US	Apple Watch	-	Paroxysmal AF	0.75
P2	Male	46	US	Apple Watch	Kardia+ ¹	Paroxysmal AF	3
P3	Female	60	US	Apple Watch	KardiaMobile ³	Paroxysmal AF	3
P4	Female	53	US	Apple Watch	Cardiogram App ¹	Paroxysmal AF	3
P5	Male	45	UK	Apple Watch	KardiaMobile ³	Paroxysmal AF	3
P6	Female	71	US	Apple Watch	KardiaMobile ³	Paroxysmal AF	2
P7	Male	72	US	Fitbit	KardiaMobile ³ , Spreadsheet	Paroxysmal AF	4
P8	Male	54	US	KardiaMobile ³	-	Persistent AF	2
P9	Male	35	US	Apple Watch	-	Paroxysmal AF	3
P10	Male	31	Singapore	Fitbit	-	Paroxysmal AF	4
P11	Female	76	US	Google Pixel	KardiaMobile ³ , Qaly App ¹	Paroxysmal AF	Undisclosed
P12	Male	45	UK	Apple Watch	Kardia+ & Support	Paroxysmal AF	1.5
P13	Male	66	Canada	Apple Watch	KardiaMobile ³ , Lookee ECG, Qaly App ¹ , Cardiogram App ¹	Paroxysmal AF	4
P14	Male	41	US	Google Pixel	-	Paroxysmal AF	1
P15	Male	33	US	KardiaMobile ³	Oura Ring	Paroxysmal AF	0.5
P16	Male	44	US	Apple Watch	Heart Analyzer App ¹	Paroxysmal AF	3
P17	Male	43	UK	Samsung Watch	-	Permanent AF	2
P18	Male	74	UK	Apple Watch	-	Paroxysmal AF	1.5-2
P19	Male	43	UK	Samsung Watch	KardiaMobile ³ , SpreadSheet	Persistent AF	0.7
P20	Female	70	Netherlands	KardiaMobile ³	Withings Watch	Paroxysmal AF	2
P21	Male	57	US	Samsung Watch	-	Paroxysmal AF	3
P22	Male	51	Norway	Withings Watch	-	Paroxysmal AF	5
P23	Male	34	US	Apple Watch	-	Paroxysmal AF	2
P24	Male	57	US	Apple Watch	-	Paroxysmal AF	4
P25	Male	65	UK	Apple Watch	Blood Pressure Monitor	Paroxysmal AF	2
P26	Male	66	Germany	Withings	Blood Pressure Monitor	Paroxysmal AF	3
P27	Male	42	US	Samsung Watch	KardiaMobile ³	Paroxysmal AF	2
P28	Female	65	US	Apple Watch	-	Paroxysmal AF	2
P29	Female	39	UK	Apple Watch	Kardia+ ¹ , TachyMon ²	Paroxysmal AF	2

Table 1: Self-Declared Participant Demographics

¹Apps with an ECG interpretation service²Customised heart rate alerting app³Portable ECG device

3.3 Data Analysis

Interview data was analysed using an inductive reflective thematic analysis (TA) approach [11, 12, 14, 16]. Following Braun and Clarke's guidelines, the first author was involved in the initial data review [15]: transcribing, becoming familiar with each interview transcript, writing reflective memos for each interview, and then reflecting upon the entire dataset [15]. The data was then analysed iteratively, with the researcher identifying patterns across the data set and inductively developing codes. Through this process, semantic codes were initially generated, for example "using technology post ablation", "using technology during AF episode", and "knowledge seeking via third party-apps". As the analysis progressed, latent codes were also generated, such as "seeking reassurance through data", "navigating high stakes self-care", "technology unchanged yet disease evolving". Throughout this process, the researcher reflected upon their research values, assumptions and experiences, with particular acknowledgement that their work in arrhythmia care in the UK's NHS and also as a personal informatics researcher undoubtedly shaped interpretation and coding [13, 16]. The researcher discussed preliminary insights and shared their reflections and codes with the research team throughout the data review phase. Upon completing the coding process, the research team collaboratively reviewed the codes and generated initial themes. The team then discussed and mapped out patterns, refined and named the final themes.

3.4 Positionality Statement

Before presenting the findings, we document our positionality and reflect on how this shaped our work. The research team is based in the UK and comprises of individuals with backgrounds in academic HCI and clinical cardiology. One author has direct experience of living with arrhythmia, while two authors have close family members living with AF. We acknowledge that our collective experiences provided certain advantages, particularly a deep understanding of the condition and its management, which participants in this study and typical users of this technology may not have.

4 Findings

Participants used multiple self-tracking tools such as wearable fitness trackers, consumer ECG devices, traditional paper diaries, spreadsheets, and mobile applications. We report on how self-tracking tools were often used simultaneously to support a broad range of self-tracking practices, focusing on: (1) learning to interpret body signals associated with AF; (2) casting a clinical gaze on their disease; (3) understanding how data and illness are experienced; (4) negotiating and shaping shared decision-making and treatment with clinicians; (5) intervening with Pill-in-Pocket for anti-arrhythmic and anticoagulant medication; (6) the complex entanglement of emotion with self-tracking technology, and finally, (7) reducing reliance on technology with disease progression. There are overlaps between some of the themes, for instance, 3) understanding how data and illness are experienced, which focuses on the sense-making process before engaging with a clinician, and 4) negotiating and shaping shared decision-making and treatment,

which focuses on participant perspectives on the role of patient-generated data in clinical encounters. These were separated for the sake of clarity for the reader.

4.1 Learning Body Signals with Technology

Participants expressed distrust in their ability to interpret symptoms, possibly linked to reduced interoception. Interoception describes how individuals perceive and regulate body signals [70]. AF patients, in particular, may experience significantly reduced interoception [59]. While reduced bodily awareness is common in chronic conditions, how interoception affects symptom reporting in chronic disease is not well understood [70].

Confirming bodily experiences with wearable ECG was a common practice. P19 said, "*I wanted objective quantification rather than just feeling it's irregular.*" Confirmation alleviated uncertainty around the participants' ability to interpret symptoms. As AF symptoms can be subtle, participants used self-collected data to learn to recognise body signals associated with AF. P7 explained, "*it gave me reassurances that I knew what was going on... when I was feeling bad, I wanted to make sure that I knew what it was.*" He explained further: "*I would pay attention to the symptoms... I never could quite get a handle on that, but I was much more comfortable having the device than trying to guess*" (P7).

This practice was seen particularly in those who had missed AF episodes due to the misinterpretation of body signals. When reviewing his historic wearable data, P1 realised he had been having AF episodes for 13 months before recognising something was wrong and seeking medical attention. He explained how challenging it was to interpret body signals after his diagnosis:

"All the things that I would have noticed I just attributed to my weight, like trouble walking up the stairs, being out of breath, the feeling overheated, the indigestion... whatever else it is like, or 'is it just in my head?' That's why I wrote it off!" (P1)

Working through the uncertainty with technology was a learning process. P5 was unaware he had AF until receiving an alert from his watch. His wearable helped him pay attention to body signals over time, "*it's trained me to recognise it [AF] now... I know that I've got it and I can tell. I can feel it and then I'll use the wearable to just check that. Yeah. To confirm my internal diagnosis as it were.*"

Despite building confidence in interpreting body signals, participants developed a reliance on technology to confirm the interpretation of body signals. P1's past experience of misinterpreting bodily experiences led him to doubt his capabilities. He explained that technology reduced uncertainty as he was able to check symptoms in real time, which was the first step in the process of learning. He explained, "*When I felt certain symptoms... I was constantly running something [ECGs] kind of seeing how's my heart behaving today versus comparing that to how I'm feeling*" (P1).

For participants living with multiple chronic conditions, it was important to understand what symptoms specifically were associated with AF or another condition. P29 was living with Long-COVID, making it difficult to discern body signals associated with AF. P29 explained, "*I already felt so unwell... I would be checking to sort of validate what was going on.*" Confirming heart rhythm

helped her understand which symptoms were associated with AF episodes.

For many, tension arose between mind and body, with anxiety often mimicking the physical experience of AF. This led participants to wonder if their perceived physical experiences were psychosomatic. P19 described how challenging he found the process of interpreting symptoms:

“I mean, I could feel it, but your mind starts playing tricks about whether you believe your own mind” (P19).

Participants used technology to understand if their perception of physical symptoms was real or psychosomatic. Rather than trusting his initial interpretation of the physical symptoms, P19 wondered if his mind had influenced his perception of physical symptoms. Taking an ECG and confirming heart rhythm took away uncertainty.

In some cases, the use of the technology did not lead to a complete interpretation of bodily experiences. P17 had not experienced anxiety in the past, but what he labelled as anxious feelings had been intermittent when he had intermittent AF and shifted to a permanent feeling of anxiousness now he was in permanent AF (*“It just seems to be an anxious feeling that’s just like ever almost, ever present”*). Whilst he used technology throughout to check physical symptoms with heart rhythm, he did not make a complete connection. Instead, he rationalised the feeling of anxiousness as something separate to his condition. Upon reflection, he recognised his anxious feelings were not associated with negative thoughts *“I’m not playing things over and over and worrying. I’m not getting up worrying!”* (P17). P14 had a similar experience, and he *“became religious about wearing [his] watch”,* using the ECG function whenever his body felt *“overwhelmed”*: *“I’m questioning now if those times that I did have panic attacks...where probably times that I actually did have afib.”* Technology provided reassurance to P17 and P14, but it did not completely take away the uncertainty in bodily experiences.

In summary, participants experienced a great deal of uncertainty around interpreting bodily experiences. Self-tracking facilitated a process of learning to recognise body signals associated with AF. However, for participants living with anxiety, it was an evolving process that could lead to dependence on technology.

4.2 Casting a Clinical Gaze

Participants were able to collect data regarding their illness in a way that mirrored some clinical monitoring. For participants such as P2, P3, P7, P19 and P29 living with symptomatic intermittent AF, capturing their AF episodes was important. P2 wanted to capture every AF episode, explaining he used his watch for *“trend watching”* and his ECG device for *“an episodic perspective”* (P2). Participants sought to monitor the stability or progression of their condition: *“I can monitor my progression or regression or all the things that are happening.”* (P4). P3 took a similar approach, capturing symptomatic AF episodes with her watch ECG as it was convenient, followed by a recording on her ECG monitor. She always took a second ECG with her ECG monitor, explaining *“the cardiologist always wants to see the data from the Kardia.”*

By capturing such ‘clinical’ data, participants were able to cast a partial clinical gaze on themselves, allowing them to monitor their condition on their own terms. P7 would monitor, intervene and track when his symptoms started: *“got into a pattern to where I*

could take a look at the Kardia [ECG Monitor] and say, ‘OK, I’m in afib’, and I would rest and then ... the Fitbit would tell me duration.” P7 recognised that his resting heart rate in AF differed from a normal rhythm rate. By monitoring his watch’s live HR, he could identify when an AF episode ended and manually log the time in a spreadsheet.

Asymptomatic participants, such as P13 and P16, relied upon their wearable to alert them to a possible AF episode, and they bypassed the Apple Watch’s recommended settings to keep alerts on despite not being FDA approved for such use: *“I have purposely not made that change [turning alerts off] because, for me, I’m asymptomatic. It doesn’t seem that I have a burden and I would much rather know the moment I’m in afib.”* (P16). AF can occur infrequently, and this can make it hard to capture during clinical tests but possible with commercial technology at home: *“The logging of the data for me, is to be able to show them what the problem is when I can’t demonstrate it and I’m in normal sinus [normal heart rhythm] for them.”* (P19).

P16’s AF episodes were asymptomatic, infrequent and short in duration, so the alert feature was instrumental in helping them understand *“the moment I’m in afib”*, but also led to constant use: *“I’m not taking it off. I want to record everything.”* Alerts were also kept on by participants who were concerned they may be getting nocturnal AF episodes. P13 kept alerts on but muted: *“I muted everything..but the alerts were there so when I woke up, I was able to catch it. I wasn’t even aware it happened, otherwise, I would have not noticed.”* P19 also adopted a similar strategy: *“Because my episodes were kicking in at night, I wanted something that would alert me when I was in afib without me having to go and check it for myself.”*

This practice not only allowed participants to monitor their condition in a clinical way but helped them build their own clinical narrative of their illness, with a considerable amount of time and effort to organise data *“a pretty rigorous log”* (P2). P1 took detailed accounts of additional arrhythmias that he was experiencing: *“I keep a log of those things and separate them into their own folder as exported PDFs”*. Each folder categorised the form of arrhythmia and was organised on a weekly basis. He did this so that he could present his data in a way that carried weight to his views during a medical appointment. Similarly, P8 took a routine approach to data collection: *“Before I open work e-mail, I do Kardia [take an ECG] just as a habit, this way I have all the data to look back on and show my cardiologist... then by the time my work day ends, I do it one more time”* (P8).

However, these logs could become complex. P19 used his ECG device and spreadsheet to *“keep a log of when it [AF] started when it ended, and therefore what the duration is,”* but the log became very complicated when he started tracking the effects of his anti-arrhythmic medication and logged other arrhythmias: *“it’s making my log a bit more difficult. I’ve added another column to say, well, what is the arrhythmia? Is it afib or is it PVCs¹ [Premature Ventricular Contraction]? Is it PVC’s plus PACs [Premature Atrial Contractions]² etc.”* Despite this, he started to realise his AF episodes were becoming

¹A premature ventricular contraction (PVC) occurs when an early electrical signal fires from the lower chamber of the heart (the ventricle).

²A premature atrial contraction (PAC) occurs when an early electrical signal originates from one of the two upper chambers of the heart (the atrium) instead of from the heart’s natural pacemaker the sinus node. This can cause a brief, irregular heartbeat.

more frequent and that he was also developing other arrhythmias explaining: “the duration is shorter thanks to the flec [flecainide an anti-arrhythmic], but the frequency is higher, presumably because the disease has progressed.” He recognised the progressiveness of the condition was rapid, which was distressing: “the increase of occurrences has been quite rapid and a bit alarming, to be honest” (P19).

In summary, self-tracking allowed participants to cast a clinical gaze upon themselves by using data from multiple devices to monitor their condition in a very clinical way outside of the clinical setting. The process helped them build a clinical narrative of their illness, which they later put to use.

4.3 Understanding Cardiac Data and Illness in Everyday Life

Participants found the process of understanding self-collected data in relation to their everyday lived experience challenging and, therefore, adopted a range of sense-making practices.

Participants recognised that understanding data alone was only the first step. It also needs to be understood within the broader experience of living with AF. To provide context, they accessed academic journals and guidelines. P8 began with “a very high-level Google search” but progressed to “a couple of scientific papers” and kept up with treatment innovations. P1 immersed himself in online resources to learn ECG interpretation and looked up medical terminology: “I would effectively look up, PAC³ [Premature Atrial Contraction], couplet⁴ or PVC⁵ [Premature Ventricular Contraction], bigeminy⁶ or trigeminy⁷” (P1). He learned to recognise his own ECG patterns by pattern matching with ECGs online that had already been interpreted, though advanced interpretation was still challenging: “Determining whether something’s from the right or the left atrium, that’s still a little bit past me” (P1).

Participants often engaged with online communities to further understanding, sometimes sharing self-collected data for interpretation. P29 explained, “I joined the community, we share our readings with each other, like look at what’s been going on today, I’ve just taken a shower and look, what I’m having to deal with.” Sharing data alongside the lived experiences of managing the condition allowed participants to contextualise the data. P13 sought others in a similar situation: “There are patients of maybe similar age as me.” P4 discussed the benefit of advice, but also the camaraderie: “the group is fantastic. They’re all kind of walking the same pathway and they offer up so much great advice.” P17 was initially nervous but found the group invaluable: “I’ve learnt the most about the condition there. If I could speak to the consultant again, I would probably have a bunch more questions for him” (P17).

In contrast, other participants turned to advanced self-tracking tools and third-party apps for AI-driven interpretation or human interpretation. These services cannot account for clinical history,

as they are not regulated ‘diagnostic’ services. P19 upgraded to a paid interpretation service after receiving ‘inconclusive’ ECGs: “instead of seeing inconclusive, it’ll tell me which of those [type of arrhythmia] it is. How useful that is? I’m not sure” (P19). P12 also struggled with advanced interpretation: “They do like long QRS and some other things I haven’t heard of.” P2 had a similar experience: “I understand what the different things mean. I get if something’s happening but don’t know why and what that means.” P11 tried several interpretation services, opting to use a human interpreter via an app, which was reassuring to them:

“It’s personalised. They express that they care, they’re sorry you’re experiencing this, they say you really should send this to your doctor. You can ask them questions. They talk about particulars that I don’t know anything about, they introduced me to things that I didn’t understand.”

Interpretation apps offered reassurance, with P29 using them when worried at night: “I’ve paid quite a few times when it’s been the middle of the night and I’ve been a bit worried.” P13 appreciated explanations of technical terms ‘PAC’ (Premature Atrial Contraction) and ‘tachycardia’ (heart rate is greater than 100 beats per minute) explaining the software can describe “what’s wrong with me and with advice saying if this is sustained for so long, you should go to emergency room things like those.” P11 and P13 became reliant on the interpretation service for shaping health decisions when “it’s not easy to get access to a doctor” (P11): “I see how serious it is and how long it lasts. And I can see some comments, advice from the software... ok ‘It’s not like life-threatening.’” (P13)

P11 used her app as a triage tool when concerned: “I probably do Qaly [app] first, to make sure I’m not wasting the doctor’s time.” The app helped her decide whether to contact her doctor. Despite not offering formal medical advice, participants still used the information to make decisions about seeking medical attention. Interpretation offered some reassurance but potentially also a false confidence. While these services offered reassurance, they did not always bring deeper understanding.

In summary, participants took different approaches to understanding their self-collected data, such as self-education, online communities, and third-party apps. Whilst self-education and engagement with online communities allowed some to build an understanding of data in relation to illness, others remained reliant on apps for reassurance. This often resulted in more questions than answers.

4.4 Negotiating and Shaping Shared Decision Making with Self Collected Data

Single lead ECGs from consumer devices are accepted evidence of AF, recognised by medical institutions [50, 55]. Participants used self-collected data to negotiate and shape shared decision-making with clinicians: “They’re the experts. I want to pick apart their expertise. But when I present an idea or what I’m feeling, I like to have some data to back it up.” (P16). Many felt data empowered them to take an active role in medical encounters: “I’m a negotiator in this, and they want us to take part.” (P11).

Some participants developed strategies for deciding which ECGs to share with clinicians. P11 used an app to filter ECG readings,

³Premature Atrial Contraction (PAC) an early heart beat which originates from the atrium.

⁴A couplet is when two early heart beats come one after the other, may originate from the atrium or ventricle.

⁵Premature Ventricular Contraction (PVC) is an early heartbeat which originates from the ventricle.

⁶Bigeminy is when every other beat is an early heartbeat, may originate from the atrium or ventricle.

⁷Trigeminy is when every third beat is an early heartbeat.

explaining *“Anything that I’m concerned about, I send to Qaly [app], and if they reassure me that it’s OK, then I don’t bother with it”* (P11). She selectively shared concerning data to avoid overwhelming her clinician. P8 initially felt he had been *“over sharing,”* learning to only share every 2-3 weeks a selection of ECGs of his own choosing: *“I don’t really have hard and fast rules, its kind of, I think this one is pretty relevant, is pretty common for what I’ve been feeling the last couple of days. So tell me if anything shows, anything crazy”*.

Participants used data to gain more influence and power in shared decision-making. For P2, it was access to the overall AF burden she was tracking using her Apple Watch: *“It helps me understand the severity overall...I think there’s a little bit more power for me to have influence over what I think is appropriate”* (P2). Access to self-collected data helped participants push for clinical intervention, viewing clinicians as gatekeepers. P1 noticed increasing amounts of arrhythmia, expediting his appointment: *“I was able to show those to my EP [specialist cardiac doctor], and kind of push. And I had those just in case, he wanted to dismiss anything and just go, ‘oh, it’s no big deal’”*. P7, P15 and P29 used data to support their preference for an ablation: *“I want solid evidence, having the watch, having the apps, I can see some data to back it up”* (P15). P29 shared her data with her doctor as it *“bolstered the argument for having the ablation”*. P14 also used data to expedite the process: *“Having the watch put in motion the steps to having the ablation.”*

Participants also used data to challenge clinical decisions. P7, after being told he was not a candidate for ablation, was initially devastated *“I didn’t want to live with this, I didn’t like the alternatives other than the ablation”* (P7). He travelled to get a second opinion, bringing his ECGs, explaining *“They could see I had over a period of 60 days I was having instances of AF every day or multiple times a day”* (P7). The second doctor agreed to do the ablation.

Conversely, some used data to support more conservative approaches to their care. P17 hoped to use his data to adjust his medications due to side effects: *“I was hoping the watch would kind of back up my argument.. and give me some stats to help justify [adjusting medications]”*. P16 cancelled his ablation procedure and used data to create a more balanced conversation with his doctor when explaining his wishes were not *“just my feeling”*: *“My viewpoint isn’t just my feeling, I’m pointing to the data. I think that helps”* (P16).

In summary, participants used consumer devices to negotiate care and shape shared decision-making. This required effort. They meticulously collected data and selectively shared data in clinical encounters to influence treatment decisions, bringing them into line with preferences, whether invasive or conservative approaches.

4.5 Intervening with Pill-in-Pocket Based on Technology

The pill-in-the-pocket approach allows patients to take anti-arrhythmic medication (typically flecainide) when an AF episode first begins. The aim is to restore normal heart rhythm and avoid an emergency admission. For participants P1, P4, P5, P12, P13, P15, P19, P20 and P22, technology had a vital role in managing their AF (see Figure 1). We found that technology was always used to confirm AF before taking medications, despite this not being part of the prescribed approach. Moreover, an emerging role of wearables may be to support a pill-in-pocket approach to blood thinning medication [91].

P1 and P4 (both based in the US) reported following this off-label approach whilst using their wearable, which is in clinical trials now [90]. P1 explained, *“I have it [anti-coagulant] there just in case, and I would rather be safe than not.”*

P20 always kept her anti-arrhythmic pills and ECG device together: *“One side has got the pill...the other has got the Kardia.”* Confirming heart rhythm was crucial before taking medication for P20: *“I take a reading and when it’s an afib, I’ve learned to take a flecainide”*. P22 took a similar approach when his body felt a *“little irregular”*: *“I did a quick ECG on my watch. I am able to take the medication I’ve been prescribed”*. P12 liked to confirm with two devices before taking the medication: *“I’ve used the Kardia or the Apple Watch both times just to confirm.”*

Despite being symptomatic, confirming brought confidence in taking the medication: *“I can feel it in my pulse when I’m in AF proper, I can just feel my wrist, and I know I’m in it, but the Kardia’s always confirmed it.”* (P12). For some, their body signals were clear, but they still always used the technology: *“I don’t even need to use my watch to be fair. But I always sort of back it up”*. Without his watch P13, felt uncertain about taking his medication despite being symptomatic: *“I have to rely on the watch...if you take my watch away now, I’ll feel hopeless. I need that all the time”* (P12). Participants relied on technology to confirm AF rather than symptoms alone. P15, aware of medication side effects, was hesitant to take medication without confirmation: *“I’m only taking it if I’m in afib and the Kardia, I rely on the Kardia to be sure..I don’t take my pulse, I check the Kardia”* (P15).

After taking the medication, participants continued using technology to monitor heart rhythm. P19 noted *“I would use my Kardia every hour”* to check. Rechecking with the device helped him determine if further intervention was required. Participants would also use the technology to reflect on the impact of their medication, and some experimented: *“The only downside I’ve heard about it is it can lead to other arrhythmias... I’m starting to get these PVCs and SVEs [also known as PAC] I wonder if that is a result of the flecainide, or was I always getting those? they [ECG] were just saying inconclusive”* (P19). P19 started to manually track dosage and outcomes, aiming to minimise medication: *“I tried upping the level and not really seen that big a difference, I then reduced again”*. While technology helped him monitor his response, inconclusive results led to doubt about the continued safe use of the medication.

In summary, participants used technology to confirm symptoms, giving them the confidence to initiate and adhere to the pill-in-pocket intervention, which was prescribed to them because they were symptomatic. Traditionally, symptoms are required to initiate the intervention rather than technology. They experimented and were able to identify if the intervention had been successful. However, when inconclusive results added uncertainty and risked misinforming participants’ decisions about whether to adhere to the medical intervention.

4.6 The Complex Entanglement of Emotion With Technology

Participants overwhelmingly felt that self-tracking offered reassurance and a sense of control, whilst living with the uncertainty of AF. P11 noted life without her watch would feel *“much less secure.”* P14

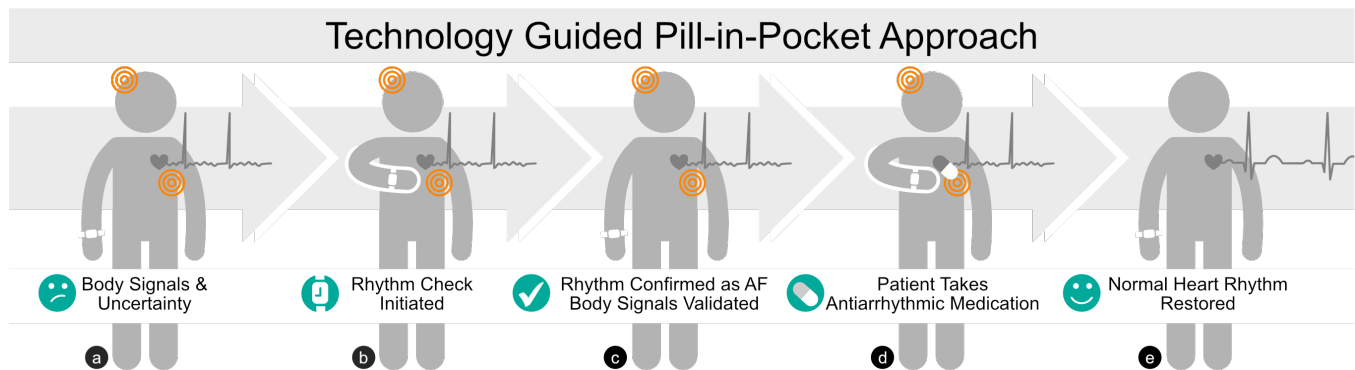


Figure 1: (a) Individual living with AF begins to experience symptoms but feels uncertain, questioning their ability to interpret the symptoms associated with AF. (b) To alleviate uncertainty, they check their heart rhythm using a wearable or portable ECG monitor. (c) If the device confirms AF. (d) They feel confident to take a pill-in-pocket anti-arrhythmic medication. (e) Within a few hours, the medication successfully converts the heart rhythm back to normal, preventing an emergency event. The individual rechecks their heart rhythm using the device to confirm the medical intervention was effective.

said it gave him “peace of mind” and made him “feel more in control”. P1 explained, “I’d feel much more anxious if I didn’t have something that I could, take a look at and verify that things are going well.” P2 was able to identify when she was at risk of an AF event based on her ECGs showing atrial ectopic beats (PACs), which provided some peace of mind: “If you know what’s gonna happen, and it does happen, that’s control... It’s reassuring” (P2).

Many felt they could not manage their condition without self-tracking. For P17, his watch allowed him to offload some of the mental burden of worrying about when an AF episode occurred: “it’s hard not to think about it all the time. My watch is keeping an eye on things for me. I suppose, at the back of my mind, it offloads a little bit of that mental pressure” (P17). P19 found that his watch reduced paranoia: “for me the data is a big upside, I would be far more paranoid if I wasn’t able to verify what my heart was doing” (P19).

However, despite this reassurance, some participants exhibited hyper-vigilance, rumination and anxiety. P17 became hyper-aware of his heart rhythm through the technology: “it’s just something that I’ve become sort of hyper-aware of”. P2 monitored in anticipation daily “I still watch every day and I expect it [AF]”. P4 admitted obsessive behaviours with the technology: “I’m a little more OCD about it than I should be”. P12 called it a “a bad habit, you use it more than to check”. P19 became preoccupied with their data when in AF: “when I’m in an episode, I can be a bit paranoid and all annoyed, am I still in it? Has it changed? And I’m checking probably too often.” Participants described periods of compulsive monitoring after initial diagnosis, with P1 taking so many ECGs that he injured himself: “I burnt my wrist running repeated ECGs. I had to switch what wrist I was using” (P1). P15 admitted that he was using it too much: “I was taking a lot, maybe even more than 30. I was taking a lot of ECG’s”.

We found that extreme hyper-vigilant periods occurred before and after medical intervention, such as cardioversion or ablation procedures. Before having an ablation, taking ECGs became a coping mechanism, helping participants prepare for an invasive and

daunting procedure: “I only use it out of a daily habit that helps me mentally prepare, this is why I’m going to go on, this is why I’m going to do a procedure that they’re going to burn part of my heart out” (P8). Before P1’s ablation, he was also “constantly checking on things, it wasn’t just even an ECG, it was looking at my heart rate”. P9, in the run-up to his ablation, was feeling “very anxious” and explained how his ECG device enabled him “to have something to fixate on”. He explained how often his wife would try and intervene “I was easily taking 5,6,7 a day she was like ‘dude you gotta chill out’. I was like ‘don’t talk to me, something feels wrong, I need to check my heart rhythm”.

After an ablation, participants felt anxious to understand if the procedure had been successful. The blanking period after ablation is a 3-month period in which the heart is healing. This period was particularly challenging for participants to navigate as patients may experience multiple arrhythmias as the heart heals: “I used it more frequently because of the blanking period” (P14). P29 experienced multiple arrhythmias in the blanking period, and it upset her when one of the results came back as AF: “I was like ‘oh no’. I was quite upset. It’s failed”. Even when she sent the results to her healthcare provider who confirmed the ECG was not AF, she still had uncertainty: “I got quite a big sense of relief. I was feeling quite down about it, but I still don’t know if I’m in the clear”.

Cardioversion procedures (electrical impulse or anti-arrhythmic medication delivered to reset heart rhythm) also caused compulsive monitoring: “I was religious. I basically walked around the house taking my Kardia readings [ECGs], trying to figure out if I was, still normal [rhythm]” (P8). The technology allowed participants to see data after these procedures that they would normally not have access to, and it could be disappointing. P8 discovered that his AF had returned: “seeing it at home by myself in this room I was like ‘ah, now that’s a bummer. And what am I going to do now?’”. P17 also discovered his cardioversion had failed just after a week. He was not aware that the procedure did not guarantee a permanent restoration of normal heart rhythm: “‘Oh, my God, I’ve got to go through that again’, it was awful.” After a number of procedures failed, P17’s

heart rhythm remained in permanent AF but he still compulsively checked his heart rhythm: *“I still look at the information. I don’t know why I still look”*.

Participants tried to cope with the emotional impact self-tracking had on them by trying to set boundaries. P1 explained, *“it’s all about managing your internal anxiety”* and he went through a process of trying to taper off how often he took an ECG a few months after his ablation when his condition was stable. P9 took a similar approach, trying to limit how many ECGs to take: *“I am trying to be conscious of not running 6 ECGs a day on my watch, which is very, very easy to do, you know, but you’re just working yourself up”*. P29 started to notice her watch was running out of battery at times since her ablation, noticing her reduced use of the watch: *“I see that as a good thing, that I’m not always aware of my heart”*.

In summary, whilst participants perceived self-tracking as offering reassurance and a sense of control over their AF, participant behaviours reflected an underlying state of hyper-vigilance and anxiety. The tension between reassurance and anxiety was particularly apparent during vulnerable periods such as after diagnosis, and pre- and post-medical intervention. In some cases, constant monitoring allowed participants to discover that medical treatments had failed, leading to profound disappointment and distress.

4.7 Reducing Reliance on Technology with Disease Progression

As participants’ AF progressed, many developed more complex arrhythmias, experienced sudden changes in heart function after procedures, or transitioned into permanent AF, which significantly influenced the usefulness of the technology.

Some participants reported experiencing more complex arrhythmias such as atrial flutter or ventricular arrhythmias (which are common in this patient group) [37, 82]. Previously useful devices struggled to track accurately with these heart rhythm changes, such as with P1, who developed atrial flutter⁸: *“I was in flutter [atrial flutter] full time, so my watch couldn’t really do much, the burden feature [% AF Burden] wasn’t doing much because I was in aflutter [atrial flutter] 100% of the time.”* P11, who developed the more dangerous ventricular tachycardia, expressed similar concerns: *“To me, that seems more dangerous than afib at this point. It’s very complicated, and the wearables cannot help you with that.”*

Following procedures like ablation or cardioversion, participants faced further challenges in interpreting their data, as these procedures caused a ‘reset’ of the heart rhythm. With little guidance on how their self-collected data might change post-procedure, many were left confused by unexpected fluctuations in heart rate and heart rate variability (HRV): *“After ablation, my HRV dropped sharply. I was surprised. What happened? Why is my HRV so low?”* (P7). Similarly, P14 described how the post-procedure changes in his data caught him *“off guard.”* The technology offered little support for participants as they navigated the recovery process leaving participants uncertain about whether changes in their data were a sign of progress or concern.

⁸Atrial Flutter is an atrial arrhythmia, which typically originates in the right atrium. Atrial fibrillation, also an atrial arrhythmia, typically originates in the left atrium close to the pulmonary veins. Atrial flutter tends to cause a faster regular heart rhythm in contrast to atrial fibrillation, which is a chaotic irregular heart rhythm. Atrial flutter requires a different treatment approach to AF.

For those who transitioned into permanent AF, the wearable device increasingly produced inconclusive results: *“In 80% of the cases I’ve got a reading [ECG], it would just tell me it was inconclusive”* (P17). P19, who experienced worsening AF, shared similar frustrations: *“I was just a bit frustrated... Why is this thing saying it’s inconclusive?”* Despite these frustrations, participants continued to use technology out of habit, even as trust in the data diminished: *“I still look at the information, despite the fact that I don’t really trust it anymore.”* (P17). Some participants developed a workaround. P17 realised if he told his wearable he was being active, his heart rate data would be more accurate: *“If I tell it I’m going on a walk, it seems to measure things accurately at that point.”* However, this workaround was found to be unsustainable, so he looked towards other technology: *“I actually resorted to using a blood pressure monitor and doing things manually to try and compare [heart rates].”* (P17). Self-tracking tools failed to keep pace with the evolving nature of the condition, leading to frustration, loss of trust, and uncertainty.

5 Discussion

In this work, we explored how consumer technology is used and experienced by individuals living with AF. In this section, we highlight how technology can shape the experience of the ill body in both beneficial and challenging ways. We will discuss three topics: managing bodily doubt, the role of the digitised heart, and the temporal nature of disease.

5.1 Managing Bodily Doubt with Technology

Some HCI research suggests that wearables can act as a ‘mediator’ and enhance bodily awareness [10, 23, 31, 97]. Boldi et al. [10] reported how wearables could enhance bodily awareness, finding that this was more pronounced in those with health conditions. In stark contrast to that work, we found participants experienced constant uncertainty about their bodies. Preferring to turn to technology to outsource bodily awareness, using ECG data to guide understanding of bodily sensations.

To better understand what our findings mean, we draw on phenomenology to introduce Carel’s concept of ‘Bodily Doubt’ [20, 21]. In contrast to doubt, bodily certainty, although not rational or logical, is taken for granted by most people in their normal daily lives when healthy and able-bodied [21]. ‘Bodily Doubt’ is where chronic illness disrupts the background feeling of implicitly knowing what our body is capable of, and ‘bodily capacity’ transitions to ‘bodily incapacity’, leading to feelings of detachment from the body (disembodiment) [20].

Carel argues that bodily doubt has three dimensions. First is loss of transparency, where previously its presence would fade into the background in everyday life; instead, the body becomes an ‘explicit object of attention’ [20]. Second, a loss of continuity: the realisation that life before illness can no longer be possible, particularly bodily habits or previous abilities [20]. Third involves a loss of faith in one’s body [20]. These dimensions all contribute to doubt, but do not necessarily explicitly manifest as an explicit bodily experience - these feelings can be implicit too. Carel describes them as a dimension of experience, a constant background feeling [20], which can come to the foreground with illness fluctuations [20].

We introduce the concept of bodily doubt because a sense of uncertainty (implicit and explicit) was found throughout our findings. Participants outsourced the interpretation of physical sensations to their wearable ECG monitors and third-party apps to mitigate bodily doubt. For some, technology helped them 'learn' to interpret bodily experiences, alleviating doubt. In these cases, the body was experienced in conjunction with technology. For others who experienced anxiety or as the disease progressed and technology failed to adapt, doubt was amplified by technology.

Using technology to navigate bodily doubt had further drawbacks, with participants turning to online peer support and unregulated apps to contextualise medical data and determine how important or unimportant a 'feeling' was in real time. The potential for misinformation or mistakes was high, and there was no accountability for misinterpretation.

Reliance on technology to navigate bodily feelings fostered a complex entanglement of emotion and technology, as participants turned to technology for control and bodily certainty [42]. However, as the disease progressed and technology introduced 'inconclusive results', use declined as it introduced uncertainty, bringing bodily doubt to the foreground. This supports other work suggesting wearable data may introduce a sense of uncertainty [4]. While self-tracking has been shown to foster a sense of control in chronic illness, our findings highlight how technology can also undermine control by exacerbating bodily doubt [3, 5, 51, 74].

Bodily doubt offers an important lens for how technology shapes the lived experience of illness, particularly the felt experience. Recent models of PI emphasise the lived experience as central to self-tracking [35, 103]. While self-tracking technology is designed to provide insight by tracking bodily experiences and encouraging reflection, our findings show how PI tools can become interpreters, not just trackers of bodily experiences, disrupting the lived experience. Beyond PI models, our findings contrast with design approaches, where "*the lived body becomes a source of design*" [52, 118]. Soma design encourages tuning in to bodily sensations, assuming users trust their ability to interpret them [52]. In chronic illness, soma design may not be possible or desired in individuals with chronic conditions who have diminished interoception and lack of trust in interpreting bodily experiences [59]. Outsourcing interpretation with technology for our participants brought confidence and reassurance.

This confidence was clear for the participants who adopted a 'pill-in-pocket' approach to medication. This approach has not traditionally involved technology and thus is only offered to symptomatic patients. However, our findings show that symptomatic participants relied heavily on technology, rather than listening to their body or even taking their pulse. This is significant as it suggests technology can support interventions in ways that the medical community has not yet explored. It shows the potential for asymptomatic individuals to benefit from this approach, as wearables can detect an AF episode they otherwise would not feel. This has implications for other 'pill in pocket' interventions, such as anti-coagulation medications, which are only being explored at the time of writing [90, 91] but were already being used by two of our participants. This requires further research to understand if technology can enhance medication adherence and outcomes in this context.

5.2 The Digitised Heart

Bodily doubt led participants to offload their felt experiences onto technology, which quantified these experiences. While the 'quantified self' has been critiqued as reductionist in HCI health research's turn to the experiential, it holds pragmatic value in clinical encounters. Clinicians prioritise data, viewing it as more objective than a patient's description of their symptoms [89]. This shift gives patients more control over their 'digitised heart', which is used as evidence in negotiating care pathways with clinicians [26, 58].

Inspired by a combination of Lupton's 'digitised body' and the 'digitally engaged patient' [72, 74], we suggest that participants formed a digital representation of their heart health, a 'digitised heart'. This was enabled by the increasing quantification and medicalisation of consumer technology. Previous work has illustrated how self-collected data helps users construct a 'layperson' narrative of illness [26]. We found that these technologies have advanced so much that they now mirror clinical measures, enabling participants to form a 'clinical narrative' [4] that allowed them to speak the "*same language*" as clinicians (P11). While barriers to using patient-generated data in clinical contexts are well documented [113, 126], our study shows that self-collected ECG data is unique in clinical encounters: it is standardised, familiar and widely accepted by cardiologists [7, 9, 77]. Participants, however, needed support in identifying abnormal ECGs such as 'atrial flutter', which differs from AF and requires a different ablation. There were several occasions where atrial flutter was coincidentally shared, leading to a change in treatment approach. Designers of the medical aspects of these consumer technologies could help users tag and prioritise data (potentially with an initial focus on atrial flutter), as participants were hesitant to overwhelm doctors.

While previous research has explored the collaborative use of self-collected health data in clinical settings [26, 80], our findings highlight a pre-clinical 'data work' [38, 39]. A phase where individuals engage in shaping their data based on their sense-making before meeting with clinicians. This process goes beyond understanding illness; it is about creating a narrative that aligns with personal health goals. Initially, individuals rely on technology due to bodily doubt, but through 'data work', they reclaim control by using the 'digitised heart' to influence clinical encounters. This 'digitised heart' allowed participants to point to something tangible and assert "*My viewpoint isn't just my feeling*" (P16). By controlling the data and deciding what to share, individuals shift some responsibility and power away from doctors.

The digitised heart was used to assert more influence in clinical encounters similar to 'boundary artefact work' [26]. However, this poses challenges as participants choose to medicalise their bodies and redefine felt experiences through unregulated technologies that lack contextual understanding of their health. This, in turn, redefines what is the lived experience of AF for these participants.

Having a digital representation of one's heart health, coupled with navigating bodily doubt raises the question: where does autonomy truly lie? Agency and autonomy have been widely explored in HCI, with it often being a design goal in self-care technologies [6, 87, 88]. It is typically framed as a trade-off between patient and clinician [88]. Our work challenges this binary view, illustrating how autonomy can dynamically shift between the user, multiple

systems (i.e. wearable, third-party apps), and the doctor. Care decisions were influenced by AI-generated ECG interpretations or third-party apps, particularly when access to professional care was limited. Our work aligns with Derboven et al. [32], who identified the role of the system as a third agent of control. According to Derboven et al. [32], this triad of autonomy may be influenced by factors such as digital literacy, perception of empowerment, and risks involved in care decisions. We extend this understanding by highlighting how participants often turned to the system using it as a triage tool before seeking professional care, or turned to the system as a stand-in when care could not be accessed (such as the middle of the night, living in remote geographical locations). This highlights the sociotechnical dynamics of how autonomy unfolds in practice.

5.3 Illness Progression and Medicalised Features Creeping into Everyday Consumer Devices

Whilst traditional self-care medical technologies have low uptake and are often criticised for not integrating into everyday life [85, 86]. Consumer devices used by participants in this study were designed to seamlessly integrate into everyday 'healthy' life with the aim of always remaining useful. However, for our participants, as the illness progressed, the use of technology decreased, although it did not result in the complete abandonment of the device itself - just its medical functions. The insights from the early stages of AF were instead replaced by constant alerts for permanent AF or, worse, inconclusive results of ECGs. This limited their usefulness for self-care at the latter stages of the illness trajectory.

We suggest that future self-care technologies need to consider a medicalised view of disease progression to anticipate future needs, but we recognise there is a design trade-off to be made between medical and holistic orientations [32]. However, we do not advocate for a purely medicalised approach. Instead, we emphasise the importance of considering the illness narrative as a design resource [41]. Technology needs to continue to provide value throughout the entire illness trajectory, but also be sensitive to the changes in people's emotional response to the data throughout. Whilst Derboven et al. [32] note the design trade-off between responsibility and reassurance in heart failure self-care, we propose that responsibility could be higher in the early stages of disease alongside offering reassurance, before switching to constant reassurance as the disease becomes advanced.

As the disease progresses, individuals may feel overwhelmed by self-tracking data [72], especially in the context in which anxiety and depression are likely to be heightened [64, 111, 123]. Our findings echo other work which recognises the need to manage anxiety alongside a desire to have full access to health data [36]. Participants preferred real-time alerts over the percentage of AF total burden as that brought more certainty. However, alerts force individuals to confront their illness when they may wish to avoid it and not deal with it if the disease trajectory is changing [72]. Designers could mitigate this with alerts that reassure users, such as notifying the end of an AF episode to reduce rumination and hyper-vigilance. Alerts could adapt and provide reassurance, highlighting stability or data trends. We found that reassurance was needed most with changes in treatment approaches and post-operative recovery.

Having access to medical-grade features respected by medical bodies is a novel development [77]. Data once only available in a cardiology clinical setting can now be obtained at a shopping mall, a scenario that would have seemed science fiction a decade ago. People's use of these devices in their everyday lives is still being understood, and in our case, participants had very good reason to be looking at medical data - this represented their heart health. Of course, with this access comes overuse and obsession. Some participants were looking at these numbers so much they had to make personal interventions to stop this 'data fetishism' [103]. One participant even burned their wrist using the ECG so much. With access to the medical side of their heart comes an emotional toll with feelings of anxiety and worry, which has similarities to work that showed the emotional labour attached to self-tracking [61]. We look again at the Soma design work. Although it prioritises the body in a way that might be at odds with our participants experiencing bodily doubt, it does echo the importance of designing for human soma across body, mind and emotion [115]

We highlight the importance of designing self-care technology for the temporal nature of disease progression and how this can be tied to supporting an individual's evolving illness narrative and emotional well-being. Whilst we propose design considerations, we are mindful that much of these will require consumer 'wellness' and 'patient education' technology to become regulated medical technology. Pandora's box of consumer medical data has been opened. There is an urgent need for regulators to catch up to the fact that people are making medical decisions from Apps which deliver empathy alongside 'educational' content when there is limited access to a cardiologist.

We believe designers of the medical aspects of these consumer devices can better support AF patients who will continue to use these technologies through their illness progression. There is an urgent need to support understanding of data and illness experience together from trustworthy sources. For example, through the design of evidence-based decision support tools that can consider patient-generated data and go beyond the one-size-fits-all approach of medical guidelines. However, in their current state, strangers online are interpreting ECGs from consumer devices without the context of medical history, akin to navigating illness without a map. The ECG may act like a compass by indicating direction, but without the map, it's hard to know if you are about to hit dangerous terrain. This highlights how self-tracking tools are leaving users unsupported and forcing them to fill in knowledge gaps on their own [110]. However, in this context, it could have very serious implications as participants used apps to decide what required urgent medical attention. Whilst sharing patient-generated data and learning from peers online has been shown to be a benefit in itself [89, 117, 125], we are concerned that online support may be influencing treatment preferences. Further work should explore how online communities and patient-generated data can influence perceptions of health and shape treatment goals.

We wish to highlight the wider implications of our work for clinical practice. AF treatment, like many other chronic conditions, is driven by the self-reporting of symptoms. There is growing concern that consumer technologies may contribute to increased medical intervention [105]. Rosman et al. [105] found that AF patients using wearables reported a higher burden of symptoms leading to higher

rates of medical intervention (i.e. ablations). Our work points toward why this may be being observed by the medical community. Our findings and those of others show that technology redefines felt experiences and enhances awareness of bodily sensations [10], but also amplifies bodily doubt and uncertainty. Future work should examine the relationship between wearables and enhanced bodily awareness/doubt, and if this leads to increased self-reporting of symptoms, which drives medical resource use.

5.4 Future Work and Limitations

Our study has several limitations. First, the sample was predominantly Western, male, well-educated, and wealthy, which limits the generalisability of our findings. As we recruited participants via an online forum, it is likely that digital literacy levels influenced participants' ability to use and experience consumer technologies. Additionally, while we aimed to recruit participants across all stages of AF disease progression, we were only able to recruit one participant with permanent AF and two participants with persistent AF. This raises the possibility that individuals with advanced AF may not be using consumer technology. Whilst our findings point towards reduced use of technology with advanced disease, we did not capture abandonment potentially because those who gave up on consumer technology would not have signed up for our study. Further research is needed to better understand if illness progression leads to abandonment in the advanced stages of AF, particularly as individuals may develop a reliance upon technology yet may experience frustration with technology as illness progresses.

Finally, the scope of our study was to explore a person's lived experience with AF and technology. It was not within our scope to explore the clinical perspective directly. Future work should explore how a pre-clinical 'data work' phase influences patient-doctor relationships and shared decision-making. We suspect that this exploration should initially focus on chronic heart conditions before being explored more broadly.

6 Conclusion

Our research contributes an empirical account of how people manage atrial fibrillation with consumer technologies through a semi-structured interview study with 29 participants. We demonstrate how consumer technology has integrated seamlessly into everyday life, with medicalised functionality creeping in, subtly redefining how we feel and manage illness. Illness can 'sneak up on us', eventually becoming part of our identity and risks 'defining us' [44]. This leads us to establish how bodily doubt and technology shape the lived experience of illness and conclude that if illness can define us, so too can technology.

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